Estudo de Caso 2 - Avaliação e comparação do retorno médio de ações

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Summary

This document provides a template for the Case Study reports. Reports should always start a short executive summary (call it an *Abstract* if you want) to give the reader a general idea of the topic under investigation, the kind of analysis performed, the results obtained, and the general recommendations of the authors.

There are (at least) two ways to render this document to **pdf** in R: one easy and one... well, less easy.

1. The easy way

Use RStudio as your editor, open the **.Rmd** file and click the *Knit PDF* button at the top of the editor.

2. The slightly-less-easy way

If you're using any other R editor (such as the basic R editor), you have to use the *render()* function from the **rmarkdown** package:

```
install.packages("devtools")                    # you only have to install it once
library(devtools)
install_github("rstudio/rmarkdown")             # you only have to install it once
library(rmarkdown)
render("report_template.Rmd","pdf_document")    # this renders the pdf
```

Experimental design

A section detailing the experimental setup. This is the place where you will define your test hypotheses, e.g.:

$$\begin{cases} H_0 : \mu = 10 \\ H_1 : \mu < 10 \end{cases}$$

including the reasons behind your choices of the value for H_0 and the directionality (or not) of H_1.

This is also the place where you should discuss (whenever necessary) your definitions of minimally relevant effects (δ^*), sample size calculations, choice of power and significance levels, and any other relevant information about specificities in your data collection procedures.

Description of the data collection

Whenever needed, you can also include an (optional) subsection describing the actual data collection, how it was performed, any adaptations or unexpected events that may have occurred, etc. Subsections like this can also be used for the sample size calculations or any other aspect that requires a longer discussion.

Exploratory Data Analysis

The first step is to load and preprocess the data. For instance, # {r loaddata} # data(mtcars) # fc<-c(2,8:11) # for (i in 1:length(fc)){mtcars[,fc[i]]<-as.factor(mtcars[,fc[i]])} # levels(mtcars$am) <- c("Automatic","Manual") #

```r
# Carregar pacotes necessários
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
# Ler o arquivo CSV com os preços de fechamento das ações
precos <- read.csv("DadosAcoesGrupoH.csv")

# Verificar os primeiros dados
head(precos)
```

```
##   X12.621 X15.372 X30.707 X9.8675 X15.301
## 1  12.340  15.537  30.039  9.8098  15.153
## 2  12.044  15.724  29.555  9.9678  14.993
## 3  12.028  15.950  29.429  9.7844  14.930
## 4  11.689  16.172  29.224  9.6984  14.948
## 5  11.490  16.156  29.124  9.6354  14.788
## 6  11.524  16.302  28.973  9.5446  14.639
```

```r
# Calcular o retorno mensal de cada ação
# Retorno mensal = (Preço atual - Preço do mês anterior) / Preço do mês anterior
retornos <- precos %>%
  mutate(across(everything(), ~ (.-lag(.)) / lag(.), .names = "retorno_{col}"))

# Remover a primeira linha com valores NA devido ao uso de lag()
retornos <- retornos[-1, ]

# Verificar os retornos calculados
head(retornos)
```

```
##   X12.621 X15.372 X30.707 X9.8675 X15.301 retorno_X12.621 retorno_X15.372
## 2  12.044  15.724  29.555  9.9678  14.993    -0.023987034    0.0120357855
## 3  12.028  15.950  29.429  9.7844  14.930    -0.001328462    0.0143729331
## 4  11.689  16.172  29.224  9.6984  14.948    -0.028184237    0.0139184953
## 5  11.490  16.156  29.124  9.6354  14.788    -0.017024553   -0.0009893643
## 6  11.524  16.302  28.973  9.5446  14.639     0.002959095    0.0090368903
## 7  11.645  16.745  29.047  9.4688  14.546     0.010499826    0.0271745798
##   retorno_X30.707 retorno_X9.8675 retorno_X15.301
## 2    -0.016112387     0.016106343    -0.010558965
## 3    -0.004263238    -0.018399246    -0.004201961
## 4    -0.006965918    -0.008789502     0.001205626
## 5    -0.003421845    -0.006495917    -0.010703773
## 6    -0.005184727    -0.009423584    -0.010075737
```

```
## 7     0.002554102    -0.007941663    -0.006352893
```

To get an initial feel for the relationships between the relevant variables of your experiment it is frequently interesting to perform some preliminary (exploratory) analysis. This is frequently referred to as *getting a feel* of your data, and can suggest procedures (such as outlier investigation or data transformations) to experienced experimenters.

```
{r pairs,fig.width=8,echo=TRUE,message=FALSE,fig.cap="Exploring
the effect of car transmission on mpg values"} # library(GGally,quietly
= T, warn.conflicts = F) # This is just me getting fancy. #
# There are much simpler ways ;-) # ggpairs(data=mtcars,columns=c(1,9),t
by transmission type", #         upper=list(combo="box"),lower=list(comb
#         diag=list(continuous="densityDiag",discrete="barDiag"))
#
```

Your preliminary analysis should be described together with the plots. In this example, two facts are immediately clear from the plots: first, **mpg** tends to correlate well with many of the other variables, most intensely with **drat** (positively) and **wt** (negatively). It is also clear that many of the variables are highly correlated (e.g., **wt** and **disp**). Second, it seems like manual transmission models present larger values of **mpg** than the automatic ones. In the next section a linear model will be fit to the data in order to investigate the significance and magnitude of this possible effect.

Statistical Analysis

Your statistical analysis should come here. This is the place where you should fit your statistical model, get the results of your significance test, your effect size estimates and confidence intervals.

```r
model<-aov(mpg~am*disp,data=mtcars)
summary(model)
```

```
##             Df Sum Sq Mean Sq F value   Pr(>F)
## am           1  405.2   405.2  47.948 1.58e-07 ***
## disp         1  420.6   420.6  49.778 1.13e-07 ***
## am:disp      1   63.7    63.7   7.537   0.0104 *
## Residuals   28  236.6     8.4
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Checking Model Assumptions

The assumptions of your test should also be validated, and possible effects of violations should also be explored.

```r
par(mfrow=c(2,2), mai=.3*c(1,1,1,1))
plot(model,pch=16,lty=1,lwd=2)
```

Conclusions and Recommendations

The discussion of your results, and the scientific/technical meaning of the effects detected, should be placed here. Always be sure to tie your results back to the original question of interest!

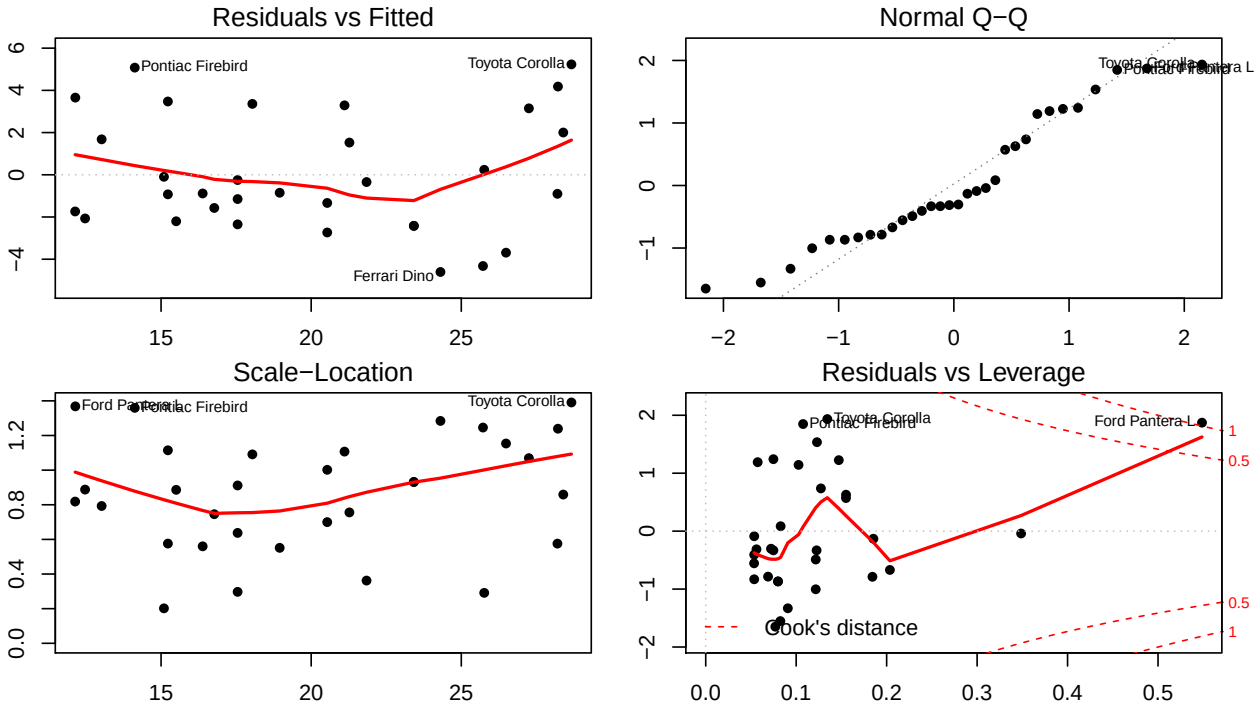

Figure 1: Residual plots for the anova model